

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? (serial link cost 64 + serial link cost 64 + Ethernet cost 1= 129)

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

 The passive interface will cause the interruption of neighbor relationships, and other routers will also be unable to learn the routing relationships between neighbors through this router.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 65 (Serial link cost 64+ Ethernet cost 1)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To allow OSPF to differentiate between high-speed links (e.g., FastEthernet, GigabitEthernet) by assigning distinct cost values, since the default reference bandwidth of 100 Mbps results in a cost of 1 for all links ≥ 100 Mbps, which does not reflect actual bandwidth differences. However, it doesn't change the route result. If the previous two paths were equivalent, they would remain equivalent after being enlarged.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

From R1 to 192.168.3.0/24, cost = $100,000/128 + 1$ (Ethernet cost) = 782

From R1 to 192.168.23.0/30, cost = $100,000/128 + 64 = 845$

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

After applying bandwidth 128 on all remaining serial interfaces, every serial link now has a cost of 781 (based on 100 Mbps reference bandwidth). The path to 192.168.23.0/30 requires two serial hops.

Thus, new cost to 192.168.23.0/24 = $781 + 781 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the new accumulated cost is 1563 after changing the cost. $1563 < 1565$, so the route to the 192.168.3.0/24 is now going through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The Router ID uniquely identifies each router in an OSPF domain. Conflicting RIDs can prevent neighbor adjacency or cause routing errors

2. Why is the DR/BDR election process not a concern in this lab?

Because all OSPF neighbor relationships in this lab are established over point-to-point serial links. This network type does not perform DR/BDR election; neighbors go directly to FULL state.

3. Why would you want to set an OSPF interface to passive?

Configuring them as passive reduces unnecessary protocol traffic, enhances security, and saves router resources.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Multi areas make the protocol hierarchical. By parting the network into a backbone area (Area 0) and multiple regular areas, it limits LSA flooding, reduces routing table size, and isolates topology changes

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be chosen because it has the lowest administrative distance.
